

NORTHGATE ACADEMY
Year 10 Combined Science - Biology Paper 1
Cell Biology, Transport and Respiration

Time allowed: 40 minutes
Total marks available: 20

Answer ALL questions in the space provided.

SECTION A - Short answer [10 marks]

- 1 Name the organelle that carries out aerobic respiration, and state the molecule it produces. [2 marks]
- 2 (a) Write the word equation for aerobic respiration. [1 mark]
- 2 (b) Explain why muscle cells contain a large number of mitochondria. [2 marks]
- 3 A cell is 50 micrometres long. In a drawing of the cell, the same cell measures 100 mm. Calculate the magnification of the drawing. Show your working. [3 marks]
- 4 Describe two ways a root hair cell is adapted for taking up mineral ions from the soil. [2 marks]

SECTION B - Extended answer

[10 marks]

- 5 State one function of the cell membrane. [1 mark]
- 6 Describe how the structure of an alveolus makes it efficient for gas exchange. [4 marks]
- 7 Define osmosis. [3 marks]
- 8 Name the product of anaerobic respiration in human muscle cells and state one effect it has on the muscle. [2 marks]

END OF PAPER